PGM — Probabilistic Graphical Models — Exam

R. Emonet – (23/01/2026) – Duration: 1h30

Student name: ....

Instructions:
- write in the most readable manner,
- give concise and clear explanations,
- do not copy the question itself,
- specify any additional assumptions you make,
- exercices are independent,
- the number of stars (★★) is an indication of the expected length for the answer (not necessarily the difficulty, and not the number of points).
- questions tagged as "*(advanced)*" are more open-ended and may require more reasoning.

Some usual formulas (if needed)

Compact notation for the Bernoulli distribution, $\mathrm{Ber}(\theta)$:

$$\text{for } x \in \{0,1\}, P(X=x) = \theta^x (1-\theta)^{1-x}$$

Gaussian/Normal 1D distribution: $\mathrm{Normal}(\mu, \sigma^2)$:

$$\text{for } x \in \mathbb{R}, P(X=x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Exercice 1 – flip flop

You play a game where the game master has a single coin and follows (without cheating) the following protocol:
- throw the coin twice
  - if the two outcomes are different, repeat (without telling the player)
  - if the two outcomes are identical, give an answer:
    - HH: give "Flip"
    - TT: give "Flop"
- NB: you do not see the coin tosses, only the answers "Flip" or "Flop"
- NB: we can use "1" for "Flip" and "0" for "Flop"

- NB: you don't know how many throws the game master needed before giving an answer

The protocol will be repeated, independently, to produce a sequence of Flip and Flop. At this point, no information is given about the rules for "winning" the game. So, we consider the sequence as potentially infinite (but fixed if look at a given time).

Possible example of observed sequences are given below (three cases, that can be used as examples in the following questions if needed, you can also suggest others):
- case 1: 00100 0101 (grouping only for readability).
- case 2: 11101.
- case 3: 11101 11011 11110 01111 10111.

Question 1 (★★)
Propose a probabilistic modelling of this game (random variables, probabilities, ...).

Question 2 (★)
Briefly discuss if the randomness in this game is aleatoric, epistemic, both.

Question 3 (★)
Under your modelling, using probabilities, provide a formal definition of each of the following problems (do not compute them here), given some observed sequence of Flip and Flop:
- what is the probability that the next observation is Flip?
- what is the probability that the next two observations are both Flip?

Question 4 (★)
Briefly explain what we call likelihood, then give the expression of the likelihood under your modelling.

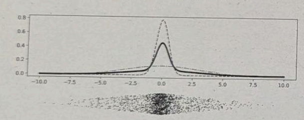
Question 5 (★★★)
Derive the maximum likelihood estimator for your modelling:
- give the steps (and tricks) you would follow to compute it from an observed sequence,
- give the derivations, using ideally an arbitrary observed sequence, or an example if needed,
- give the closed-form solution, if it exists.

Question 6 (★) – (★★★★) *(advanced)*
At some point after we observed a sequence, I propose you a betting game: you win 100€ if the next two observations are both Flip, else I take the money you bet. We are interested in t (for "threshold"), i.e., of how much you should bet (at most):
- provide a formal definition of the question, using probabilities,
- derive the answer under your modelling.

Question 7 (★★) *(advanced)*
We now consider that we agreed on the following rules for a different game:
- we each put 50€ in the pot,
- we wait until we get a sequence of length 11,
- you win the 100€ if there is a majority of Flip (6 or more) in the final sequence, else I take the 100€.

We start the game, but after we got 5 observations (e.g. the one from case 2: 11101), the game is interrupted. I propose to split the money 50/50:
- would you accept it (is this proposal fair to your)?
- if not, discuss/propose a better split that I would also accept (justify your answer)

Exercice 2 – noisiness



We consider a situation where a device provides noisy values (details below). The goal is to understand the noise characteristics so that, later, someone can use it to better interpret measurements from the device.

The experts conduct a controlled calibration experiment where they know the true value (e.g. from an expensive noise-free device), and they collect noisy measurements. Thanks to this experiment, we obtain observed noise values (the difference between the measured value and the true value) that we call x.

Domain experts tell us that:
- there are two exclusive noise sources,
- each measurement is affected by exactly one of them,

- each noise type follows a normal distribution centered at 0 with different standard deviations,
- the standard deviation of the first type of noise is smaller, i.e., type 1 is "low" noise, type 2 is "high" noise,

The experts do not know the relative frequency of each noise type, nor their actual variances, they want to find that information. Beyond the above points (which is a strong prior on the overall process, that we suppose correct), the experts provide no further priors.

If necessary, we suppose $N = 1000$ noise values are observed in this controlled experiments.

Question 8 (★★)
Provide a probabilistic modelling of this problem (random variables, probabilities, ...).

Question 9 (★)
If you didn't already specify it in the previous question, explicit the number of variables that are present in your model, whether they are observed or not, whether they are "parameters" (i.e. interesting for the experts).

Question 10 (★★)
Explain the big-picture of the Expectation Maximization (EM) algorithm applied to this problem.

Question 11 (★★★)
Derive the EM steps for this problem.

Question 12 (★★)
Explain the big-picture of variational inference (VI) applied to this problem, focusing of the similarities and differences with EM (do not derive the VI equations).

Question 13 (★★)
How would the principle of "amortization" could be applied to this problem?

Question 14 (★★★) *(advanced)*
Provide the details of what VI is minimizing and derive some VI equations for this problem.

(End of exam)